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Towards Robust Driving Perception: A Flexible Scale-Driven Family for Self-Supervised Monocular Depth Estimation

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Project Page & Demo Videos
startnew.github.io/projects/flexdepth
Code: github.com/startnew/flexdepth

Motivation

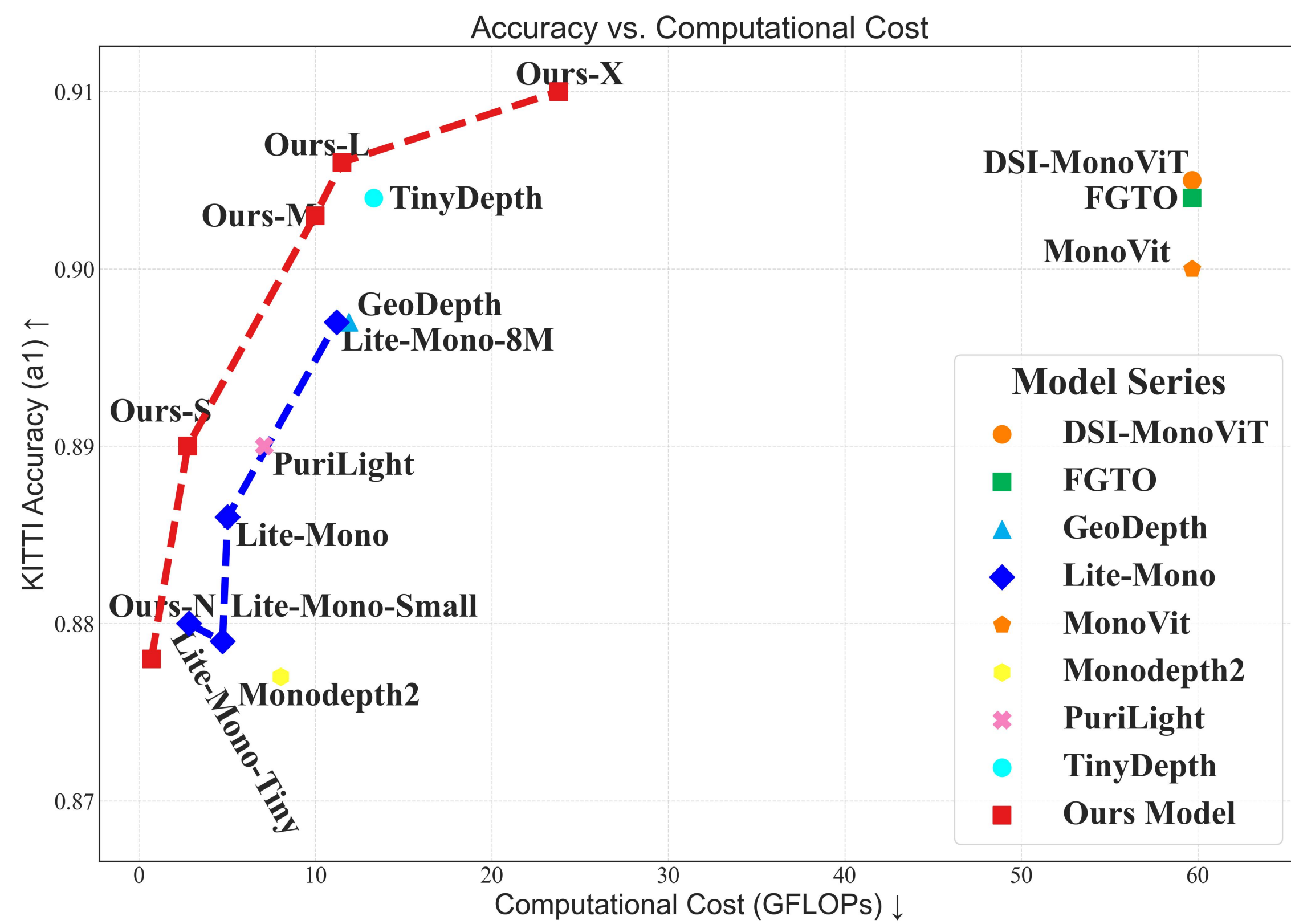
- Self-supervised monocular depth models are **locked to a single scale** and degrade sharply in dynamic driving scenes.
- Dynamic-scene-specialized networks are **too heavy for edge devices** in vehicles.

→ **One model family: every scale, robust to dynamics.**

Contributions

- FlexDepth** — a scale-driven lightweight self-supervised depth **family** (Nano → X-Large) studying the accuracy–efficiency trade-off.
- Two-stage static–dynamic decoupled training** — independent confidence evaluation for static background and dynamic objects.
- Scale-Driven Decoder (SDD)** — efficient inference + accurate depth, elastically configurable per device.

Accuracy vs. Compute — KITTI



FlexDepth (red) dominates the accuracy–compute frontier at every scale, from 0.7 GFLOPs (Nano) to 24.6 GFLOPs (X-Large).

Model Family & Efficiency

Model	Params	GFLOPs	4090	V100	SD8Elite
Flex-Nano	1.5M	0.7	1583	774	37.6
Flex-Small	6.1M	2.8	1215	606	18.6
Flex-Medium	12.7M	10.0	510	231	5.8
Flex-Large	15.2M	11.5	467	216	5.2
Flex-X-Large	32.3M	24.6	321	130	3.0
Lite-Mono-Tiny	2.1M	2.8	1048	312	15.9
MonoViT	33.6M	59.7	265	126	1.9

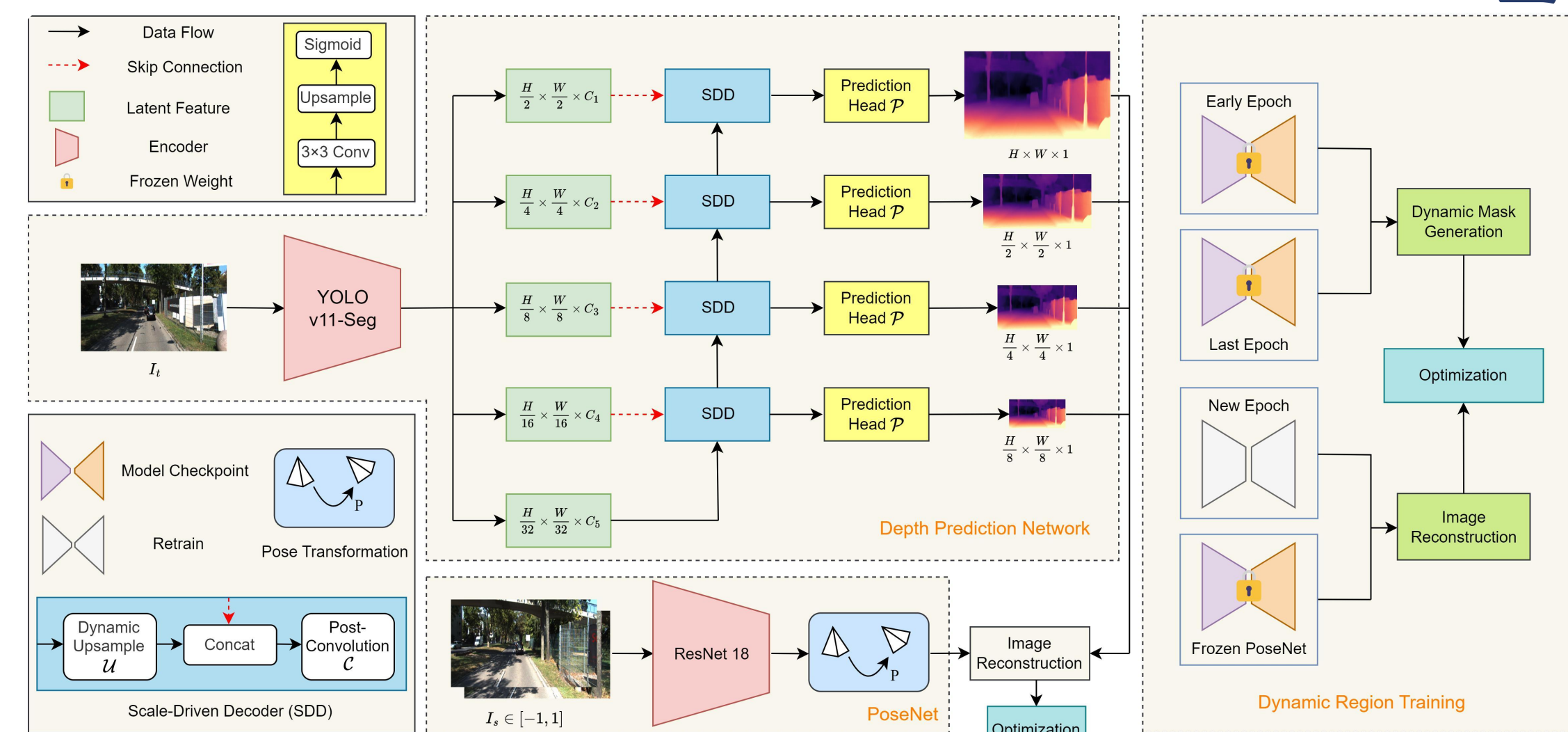
640×192 input; FPS: GPU: batch 16; Snapdragon 8 Elite: batch 1, performance mode.

Zero-Shot Generalization — Make3D

Method	Abs Rel↓	Sq Rel↓	RMSE↓	RMSE log↓
Monodepth2	0.322	3.589	7.417	0.163
GeoDepth	0.296	2.750	6.735	0.153
Flex-X-Large (Ours)	0.279	2.799	6.599	0.144

KITTI-trained models tested directly on Make3D — no fine-tuning.

FlexDepth Overview — Two-Stage Training



Stage 1: depth network (YOLOv11-Seg encoder + SDD) and PoseNet are trained jointly with photometric self-supervision. Stage 2: motion-induced inconsistency between early- and late-epoch predictions yields dynamic-object masks; with a frozen PoseNet the depth network is retrained — zero extra inference cost.

Scale-Driven Decoder (SDD)

- No pre-convolution** — global semantics are preserved before spatial expansion (removes the bottleneck of classic upsample–concat pipelines).
- Dynamic upsampling** — learned content-aware offsets + Pixel Shuffle + grid sample give sharp, non-uniform upsampling and crisp object boundaries.
- Two post-conv bottlenecks** — HEB (dense, short gradient paths) for small models; HPB (CSP-style, fine-grained) for large ones.
- Inverted prediction head** (upsample → predict) for larger models; classic head keeps Nano/Small real-time.

Decoder Configuration per Family Member

Family member	Post-Conv C	Upsampling U	Head P
Nano / Small	HEB	Dynamic	Classic
Medium / Large / X-Large	HPB	Dynamic	Inverted

Static–Dynamic Decoupled Training

- Stage 1** — photometric loss (SSIM + L1) with minimum reprojection & auto-masking, edge-aware smoothness; depth net + PoseNet (ResNet-18, 6-DoF) trained jointly.
- Stage 2** — dynamic regions betray themselves: early- vs late-epoch disparity disagrees. Mask M uses an **adaptive per-pixel threshold**; retrain with masked normal-distribution loss + geometric depth smoothness.
- No auxiliary signals** — no semantics, no optical flow, no pseudo depth; single frame at test time.

$$M = [|\text{disp}^E - \text{disp}^L| < g(\varphi(f_s^E, f_s^L))]$$

$\beta = 0.7$; $g(\varphi(\cdot))$: adaptive per-pixel threshold. Static where early/late-epoch predictions agree; dynamic otherwise.

Training Recipe

- NAdam optimizer**. Epochs (Stage 1 + Stage 2): KITTI 30 + 20, Cityscapes 30 + 10.
- Batch 12** (Nano/Small; encoder lr 5e-5, decoder 1e-4), **batch 6** (Medium and above; lr 5e-5).
- Flex-X-Large trains on KITTI in **~10 h on a single RTX 4090**.
- Monocular videos only** — KITTI Eigen split & Cityscapes; no extra labels.

Results — KITTI (Eigen Split, 640×192)

Method	Abs Rel↓	Sq Rel↓	RMSE↓	$\delta < 1.25\%$	GFLOPs↓
Monodepth2 (19)	0.115	0.903	4.863	0.877	8.0
Lite-Mono-Tiny (23)	0.110	0.837	4.710	0.880	2.8
PuriLight (26)	0.106	0.747	4.536	0.890	7.1
Flex-Nano (Ours)	0.110	0.794	4.678	0.878	0.7
Flex-Small (Ours)	0.104	0.713	4.458	0.890	2.8
GeoDepth (25)	0.100	0.694	4.381	0.897	11.9
TinyDepth (24)	0.096	0.665	4.249	0.904	13.3
Flex-Medium (Ours)	0.096	0.639	4.253	0.903	10.0
Flex-Large (Ours)	0.095	0.642	4.199	0.906	11.5
MonoViT (22)	0.099	0.708	4.372	0.900	59.7
DSI-MonoViT (25)	0.096	0.711	4.321	0.905	59.7
Self-Distillation (25)	0.095	0.619	4.156	0.905	64.5
Flex-X-Large (Ours)	0.093	0.605	4.114	0.910	24.6

Results — Cityscapes (416×128)

Method	Abs Rel↓	Sq Rel↓	RMSE↓	$\delta < 1.25\%$	GFLOPs↓
ManyDepth (21) 2f	0.114	1.193	6.223	0.875	12.1
DynamicDepth (22) 2f+5e	0.103	1.000	5.867	0.895	18.5+
PuriLight (26)	0.099	1.014	5.892	0.901	5.7
Flex-Nano (Ours)	0.107	1.261	6.133	0.893	0.6
Flex-Small (Ours)	0.100	1.078	5.813	0.904	2.2
Flex-Medium (Ours)	0.089	0.885	5.358	0.917	8.0
Flex-Large (Ours)	0.087	0.911	5.310	0.924	9.2
ProDepth (24) 2f	0.095	0.876	5.531	0.908	35.5
DSI-MonoViT (25)	0.088	0.872	5.410	0.918	47.7
Flex-X-Large (Ours)	0.086	0.877	5.268	0.926	19.6

M: monocular video only. 2f: two test frames. +5e: extra semantic segmentation.

Dynamic-Region Evaluation — Cityscapes

Method	Aux.	Dyn AbsRel↓	Dyn $\delta\%$	Sta AbsRel↓	Sta $\delta\%$	GFLOPs↓
ProDepth	–	0.115	0.884	0.092	0.911	35.5
Mining-BiNet	✓	0.119	0.872	0.080	0.922	27.0
DSI-MonoViT	–	0.097	0.918	0.086	0.919	47.7
Flex-X-Large (Ours)	–	0.091	0.935	0.085	0.925	19.6

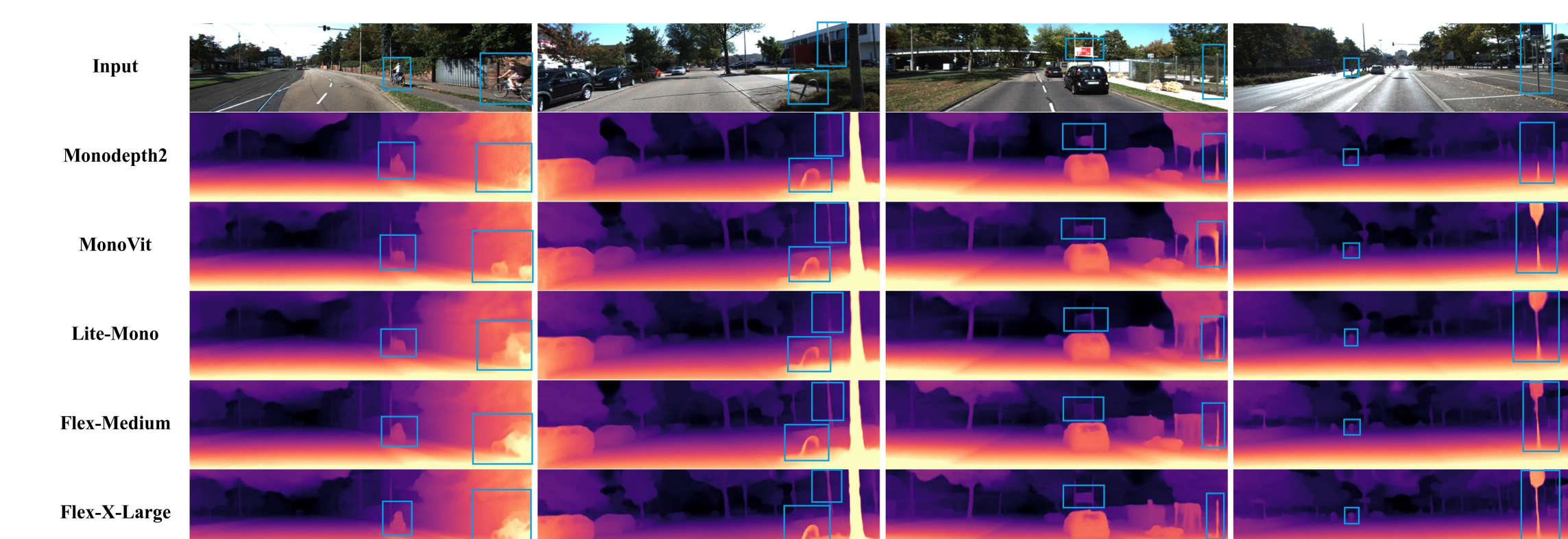
Aux.: pixel motion / pseudo depth. Best dynamic-region accuracy with the lowest compute — using no auxiliary cues.

vs. Foundation Models (Depth Anything V2)

Method	Type	Params	GFLOPs	AbsRel↓	$\delta < 1.25\%$
DA2 ViT-L @1722×518	Zero-shot	335M	1947	0.070	0.956
DA2 ViT-S @1722×518	Zero-shot	25M	137	0.077	0.944
DA2 ViT-L @644×196	Zero-shot	335M	276	0.092	0.915
DA2 ViT-S @644×196	Zero-shot	25M	19	0.110	0.881
Flex-X-Large† (Ours)	Self-sup.	32M	25	0.063	0.952

KITTI, LS alignment + improved GT, † ~1/10 params & ~1/78 FLOPs of DA2 ViT-L; yet lower Abs Rel (0.063 vs 0.070).

Qualitative — KITTI

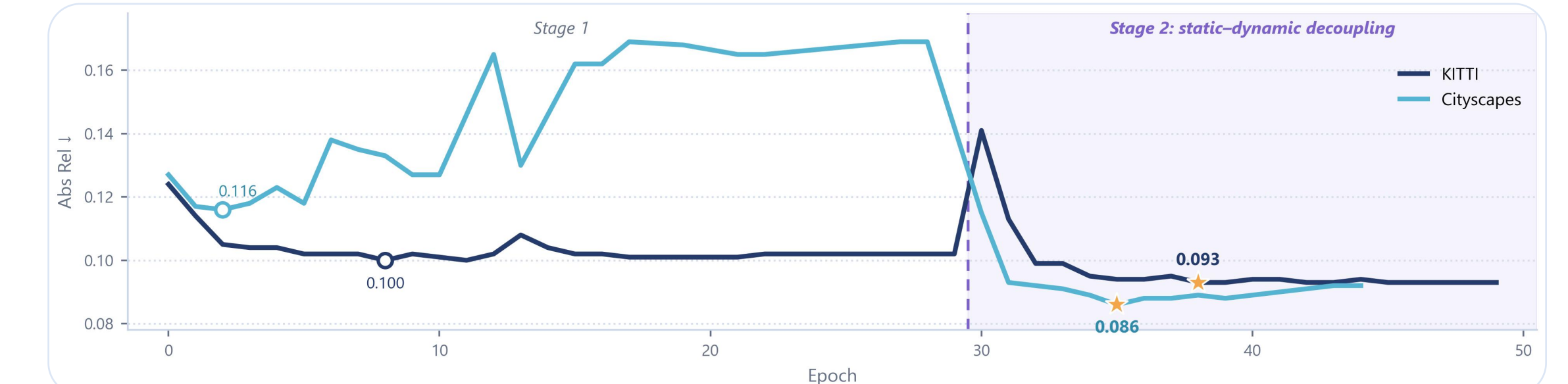


vs. Monodepth2 / MonoViT / Lite-Mono: sharper thin structures and more complete objects (blue boxes).

Dynamic Objects Across Training

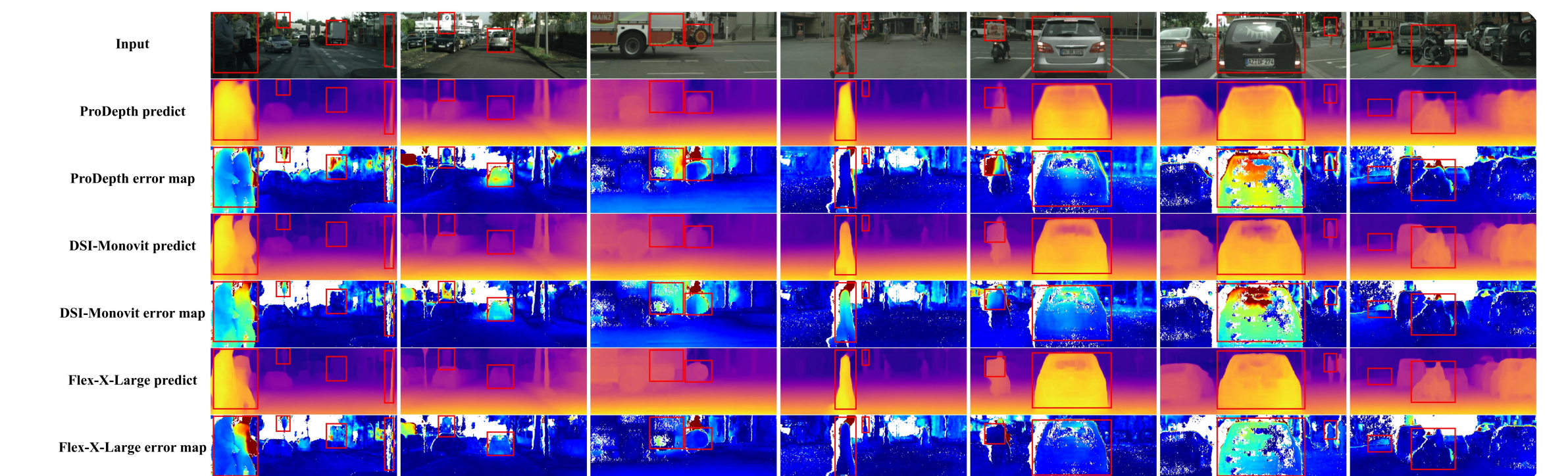


A customized μ threshold per scene yields precise dynamic-region masks across varying environments (black = dynamic).



Stage-2 static–dynamic decoupling (shaded) further reduces Abs Rel on both datasets — KITTI 0.100 → 0.093, Cityscapes 0.116 → 0.086.

Qualitative — Cityscapes (Dynamic Scenes)



Depth & error maps (blue = small, red = large error) vs. ProDepth and DSI-MonoViT. Flex-X-Large is cleanest on dynamic objects (red boxes).

Ablation Highlights (Flex-X-Large, KITTI)

- Full model: 0.093 Abs Rel / 0.910 δ** — best overall.
- w/o Stage-2 decoupling: 0.100 / 0.900** — largest drop.
- Gain ≠ more training:** Stage-1 best = 0.100 vs. Stage-1+2 = 0.093.
- Adaptive mask > fixed threshold:** Sq Rel 0.605 vs. 0.629.

Conclusion

- FlexDepth** — a flexible scale-driven self-supervised depth family for robust driving perception.
- SOTA at every scale** on KITTI & Cityscapes — no auxiliary info, single frame.
- Strong on dynamics** (CS dynamic $\delta = 0.935$) and zero-shot (Make3D Abs Rel 0.279).
- Edge-ready:** Flex-Nano 0.7 GFLOPs → 37.6 FPS on Snapdragon 8 Elite.

Acknowledgement & Links

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Code: github.com/startnew/flexdepth

Project & demo videos: startnew.github.io/projects/flexdepth